CASE STUDY:

Computation of PI, area of circle & sphere using Montecarlo method:

This section presents the case study, in which PI, area of circle & sphere has been computed. The program is written in Python. This case study includes the use of following concepts;

1. File IO.
2. NumPy array.
3. Random Number Generator.
4. Montecarlo method.
5. MPI for parallelization of code.
6. Collective Communication (Broadcast and reduce).

Montecarlo method is a statistical technique used to estimate the numerical quantities using random sampling. In this case study, Montecarlo method is used to estimate following quantities;

1. The value of Pi.
2. The area of Circle.
3. The volume of Sphere.

Montecarlo method works by generating random points inside a known geometric region and estimate the quantity by determining the fraction of points that fall inside the target shape.

$$\text{Probability} = \frac{\text{Points inside the target shape}}{\text{Total points}}$$

For this case study, it is considered that circle of radius R is inscribed in a square of area equal 1. The random points are generated inside the square, the probability that random points fall inside the circle is given by;

$$\frac{\text{Points inside the target shape}}{\text{Total points}} = \frac{\text{Area of Circle}}{\text{Area ofSquare}} = \frac{\pi\,(0.5)^2}{1} = \frac{\pi}{4}$$

Where Pi can be estimated as;

$$\pi_{estimate} = 4\,\frac{\text{Points inside circle}}{\text{Total points}}$$

For area of circle;

Area of circle = $\pi_{estimate}\ r^2$

In the same way this can be done for sphere, volume of sphere is given by;

$$V = \frac{4}{3}\pi r^3$$

Where sphere is enclosed inside a cube of volume equals to 1.

$$\frac{\text{Points inside the target shape}}{\text{Total points}} = \frac{\text{Volume of sphere}}{\text{Volume of Cube}} = \frac{\pi}{6}$$

Where Pi can be estimated as;

$$\pi_{estimate} = 6\ \frac{\text{Points inside circle}}{\text{Total points}}$$

Volume of sphere = $\frac{4}{3}\pi_{estimate}\ r^3$

To start, program reads the input file for initial data:

"Total points": 1000000,

"dimensions": 2,

"radius": 0.5

Once the total number of points are read, the root process, usually process 0 send this data to all the process in communicator using the **broadcast** operation. The idea is to divide total number of points in N process, so that each process uses Montecarlo method to see how many points lie inside the circle. The program is written to handle the situation where number point distributed among process may not be equal.

The point is considered to be inside circle if;

$(x - 0.5)^2 + (y - 0.5)^2 \leq 0.5^2$

Each process computer the number of points lie inside circle and send this data to root using **Reduce** which is collective communication mpi operation. It sums up all the data and final result is received in root process. Root process then estimates Pi, if the estimated Pi converge to a threshold value of 1.0e-5, the value is considered and area of circle is then computed, otherwise root process increments number of points and broadcast it again all the process. Then the process repeats itself until the value of estimated Pi is converged.

The same process is followed for Pi estimate and volume of sphere, with only difference that dimension variable is now equals 3.

At the end of process, the results are written in the output file along with the computation time.

The program is run with processes 2, 4, 8 ,16, 20.

The effect of strong scaling is also presented in the study.